Ashutosh Tiwari

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Work Experience

Senior Machine Learning Engineer (ML Platform & Frameworks) — Jul. 2024 – Present
ADOBE FIREFLY — *GENERATIVE AI, COMPUTER VISION, LARGE LANGUAGE MODELS*

- Lead fault-tolerant distributed inference and training infrastructure for the org's ML platform: GPU pipelines on Kubernetes engineered to survive node loss and partial failure, from hyperscaled daily EMR ingestion through production model inference.
- **Distributed Inference Frameworks**: Designed and led engineering for the org's online and offline inference systems powering experimentation and production at scale, reaching 1000s$\times$GPUs per model job across many model queues in one parallel wave.
  - Built online inference on vLLM + Ray for rapid experimentation with LLMs such as Falcon-40B and Llama 2 70B.
  - Built offline inference on Ray Data + PyTorch Lightning; a module benchmarked on that engine at 1.09M rows, 8$\times$A100, 16 fractional-GPU actors. Introduced torch.compile(mode="max-autotune") in production: compiler-generated Triton kernels, fusion, CUDA-graph capture.
  - Modeled pipelines as horizontally-scaling Source $\to$ Transform $\to$ Sink plugins with no pod-to-pod communication, KEDA-autoscaled on SQS queue depth; batched sends retry on exponential backoff with full jitter, and undeliverable work drains to a DLQ.
- **Foundation Model Training Framework**: Leading development of a PyTorch-based training framework adopted across the org for training generative models on billions of images and videos.
  - Designed a plugin-based strategy pattern over FSDP/FSDP2 to make distributed training strategies pluggable.
  - Integrated Flash Attention 2/3, context parallelism, and distributed attention for DiT text-to-image and text-to-video training.
  - Grew the engineering bench: mentored engineers onto the framework, authored its onboarding guides, and reviewed their integrations.
- **Build & Deployment System**: Built a CLI-driven build system (BuildKit images on KEDA-scaled Kubernetes jobs, pushed to ECR), fronted by a Rust control plane and an MCP interface for AI-agent-driven builds, deployments, and inference.
- **Enrichment Service**: Orchestration over the inference framework; ran fire-and-forget pipelines behind a concurrency-safe S3 checkpoint cache, lease/TTL heartbeats and a circuit breaker.
- **Training-Data Governance**: Solely authored the org's fault-tolerant training-data lineage subsystem in Rust: idempotent claim/receipt registration, immutable S3 evidence layouts, and manifest-after-data publication.
- **Data Quality Framework**: Wrote the org's first data quality framework on Cerberus, used across enrichment pipelines to validate the correctness of feature-generation outputs across a Billion asset catalog growing 35–38M images/month.

Software Dev Engineer II (ML Platform) — Jan. 2019 – Jul. 2021
SWIGGY — *DATA SCIENCE PLATFORM, TIME SERIES FORECASTING*

- **Online Inference Platform**: Core contributor to the platform that served real-time model inference to 18M+ monthly transacting users across food delivery and quick commerce, autoscaled on live load metrics.
  - Engineered the dynamic request-routing layer that matched incoming API calls against geo-tags, traffic profile and service metadata, then dispatched them to model clusters partitioned by region and load domain.
  - Implemented routing and dispatch on an Akka-style actor framework: resilient asynchronous invocation, fault isolation across clusters, and backpressure-aware scheduling that held sustained low P99 under high QPS.
- **Feature Store**: Built and maintained the feature store and its pipelines, serving on-demand features to production ML models at 4Bn rows and 10K QPS, with multichannel ingestion via Spark, Flink, and user files.
- **Forecasting & Correlation Platform**: Founding member of the time-series forecasting platform adopted by teams across the org; its forecasts drove critical scaling decisions in real time.
- **DAQ**: Led DAQ, a large-scale API scraping tool collecting 15M rows daily for analysis and model training.

Software Development Engineer (Search Relevance) — Sep. 2017 – Jan. 2019
FLIPKART (a Walmart company) — *SEARCH RELEVANCE, QUERY INTENT, NLP*

- **Search Intent Models**: Built and improved CRF and neural-network intent models powering search and discovery for millions every day; drove the error-class analysis, designed the fixes and shipped them.
- **Tail Query Classifier**: Shipped a FastText-based query-store classifier that predicts the category of a tail query.
- **ML Workflow Automation**: Built the first workflow at Flipkart to retire manual deploys, automating training and auto-deployment of search models—written first in Luigi and later migrated to Airflow.
  - Wrote a generic Airflow framework that builds DAGs at runtime per ML model and orchestrates the train$\to$deploy flow, gating every auto-deploy on data and model validations.
- **Large-Scale Data Pipelines**: Implemented Cascading/HDFS pipelines over 4Bn+ user-event datapoints, extracting and transforming them into the training data the search models learned from.

Software Development Engineer — Sep. 2016 – Sep. 2017
GROUPON — *BACKEND ENGINEERING, RECOMMENDATION SYSTEMS*

- *Customer segmentation and deal-recommendation ML (NSVD unsupervised collaborative filtering on Neo4j); Cyclops customer-service platform live in every Groupon country.*

Software Engineer — Sep. 2015 – Aug. 2016
NETSPEED SYSTEMS (Acquired by Intel) — *GRAPH ALGORITHMS, NETWORK ON CHIP*

- *C++ graph algorithms for Network-on-Chip interconnects: Virtual Channel Arbitration, Polarity-based Arbitration, Multi-Cast Filtering, Structural Latency Breakdown.*

Publications

Accepted at *NetSci 2023* (Poster Presentation) and *IC2S2 2023* (Parallel Talk) as **first author**

[1] **Ashutosh Tiwari**, Prof. Sadamori Kojaku, Prof. Yong-Yeol Ahn, "Biased Contrastive Learning debiases Graph Neural Networks," *International Conference on Network Science (NetSci)*, 2023. Poster

A non-parametric contrastive framework for fairness-aware graph representation learning: it debiases GNN embeddings with respect to sensitive node attributes and structural homophily for more organic recommendations.

Education

Indiana University
Master of Science in Computational Data Science — *Aug. 2021 – May 2023*

National Institute of Technology
Bachelor of Science in Computer Science — *Aug. 2011 – May 2015*

Research Experience / Independent Study

- Debiasing independent study at IUNI (Profs. YY Ahn & S Kojaku) on fair recommendation from biased graphs; paid RA on **"User Intent as a Network"** (Kelly Business School); NLP Lab@IUB with Prof. D Cavar on TieML and event-timeline modelling with fine-tuned LLMs.

Teaching Experience

- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2023
- Teaching Assistant for Machine Learning (CSCI-B 555) with Prof. R Khardon — Fall 2022
- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2022

Selected Projects

BiasNet | *Deep Reinforcement Learning, PyTorch, Actor Critic Algorithm, On-policy Model Free* — Code/Report
- Learning to fight in Street Fighter II with induced relational bias from differential game scenes.

DeepFoodie | *Python, Tensorflow, Self Supervised Deep Clustering, Deep Learning, Transfer Learning* — Code/Report
- Clustering dishes on basis of their ingredient embeddings. These ingredient embeddings are generated by a NN.

BlindNet | *Python, Pytorch, Deep Learning, Transfer Learning* — Code/Report
- Image to vector generation on Coco dataset.

Continuous Dominant Set Repair | *C++, Graph Algorithms, Guha and Khuller's Algo., Greedy* — Code
- Repairs a broken link in Continuous Dominant Set in $O(\Delta^2)$, where Δ being the avg cardinality of connected graph.

Humana Mays Healthcare Analytics Case Competition | *Boosted Trees, Feature Engineering* — Code
- 11th rank on leaderboard. Hosted by TAMU and Humana Mays, 2021.

AnalyticsVidhya Solutions | *Python, Tensorflow, Torch, Time Series, Feature Engineering, Catboost* — Code

Investigating Bias Manifolds | *Bias Manifolds, Bias Progression, Measuring Bias, Python, Word2vec* — Code/Report

Technical Skills

Search Relevance, Recommendations, Graph Neural Networks, Fairness Aware Modeling, Computational ML, NLP, Computer Vision, DL, Spark, Python, Scala, C++, Contrastive Learning, Large Language Models
Frameworks/Libraries/Tools: Ray, Pytorch Geometric, Pytorch Lightning, Pytorch, Tensorflow, Sklearn, Numpy, Pandas, Matplotlib, HDFS, Spark, Kafka, Flink, AWS Sagemaker, DynamoDB, Faiss, vLLM, Django